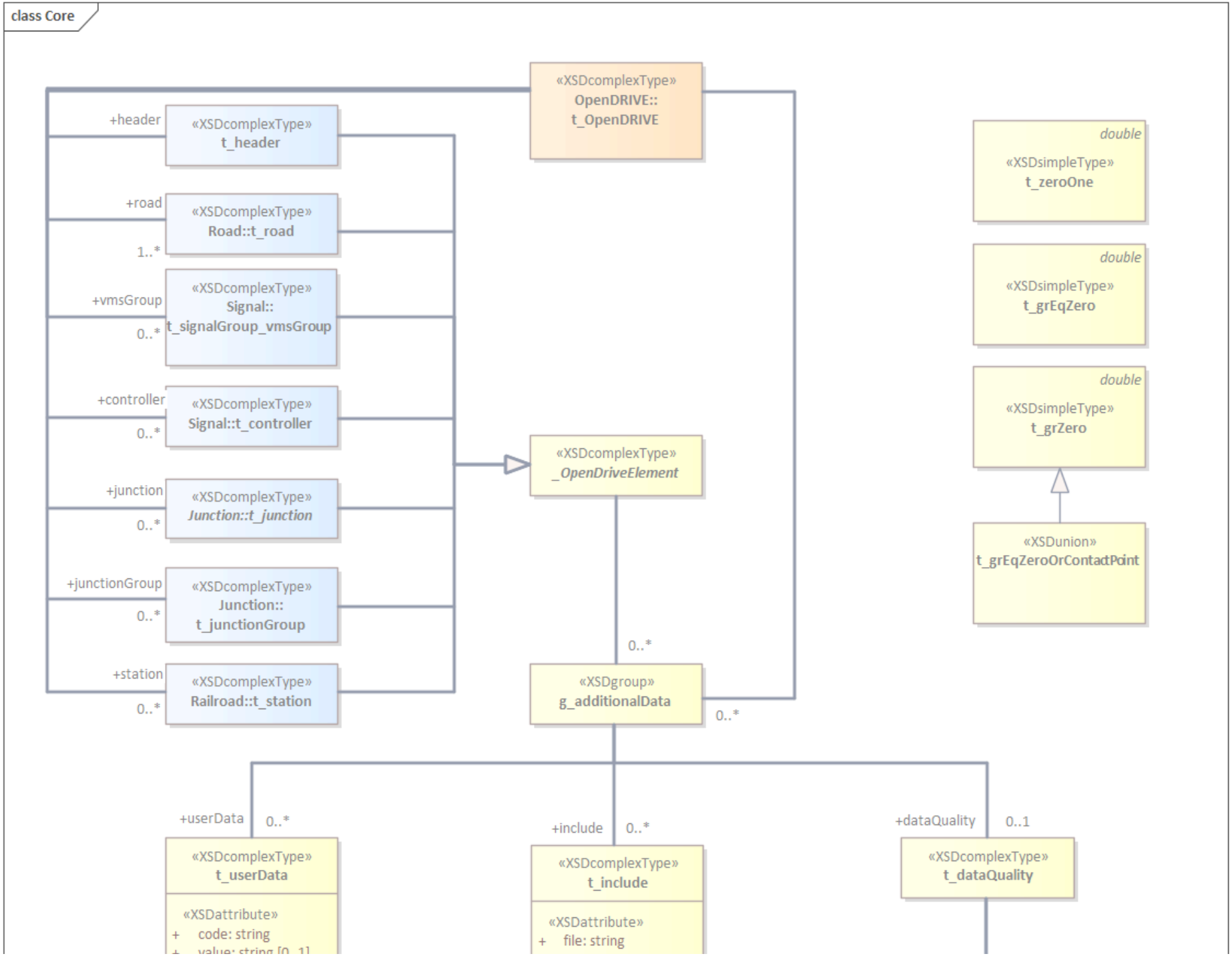
OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

7 Additional data

7.1 Introduction

ASAM OpenDRIVE offers the possibility to include external data. The processing of this data depends on the application. Additional data may be placed at any position in ASAM OpenDRIVE.


class Core
«XSDcomplexType»
OpenDRIVE::
t_OpenDRIVE
+header
«XSDcomplexType»
t_header
+road
«XSDcomplexType»
Road::t_road
1..*
+vmsGroup
«XSDcomplexType»
Signal::
t_signalGroup_vmsGroup
0..*
+controller
«XSDcomplexType»
Signal::t_controller
0..*
+junction
«XSDcomplexType»
Junction::t_junction
0..*
+junctionGroup
«XSDcomplexType»
Junction::
t_junctionGroup
0..*
+station
«XSDcomplexType»
Railroad::t_station
0..*
«XSDcomplexType»
_OpenDriveElement
0..*
«XSDgroup»
g_additionalData
0..*
double
«XSDsimpleType»
t_zeroOne
double
«XSDsimpleType»
t_grEqZero
double
«XSDsimpleType»
t_grZero
«XSDunion»
t_grEqZeroOrContactPoint
+userData
0..*
«XSDcomplexType»
t_userData
«XSDattribute»
+ code: string
+ value: string [0..1]
+include
0..*
«XSDcomplexType»
t_include
«XSDattribute»
+ file: string
+dataQuality
0..1
«XSDcomplexType»
t_dataQuality

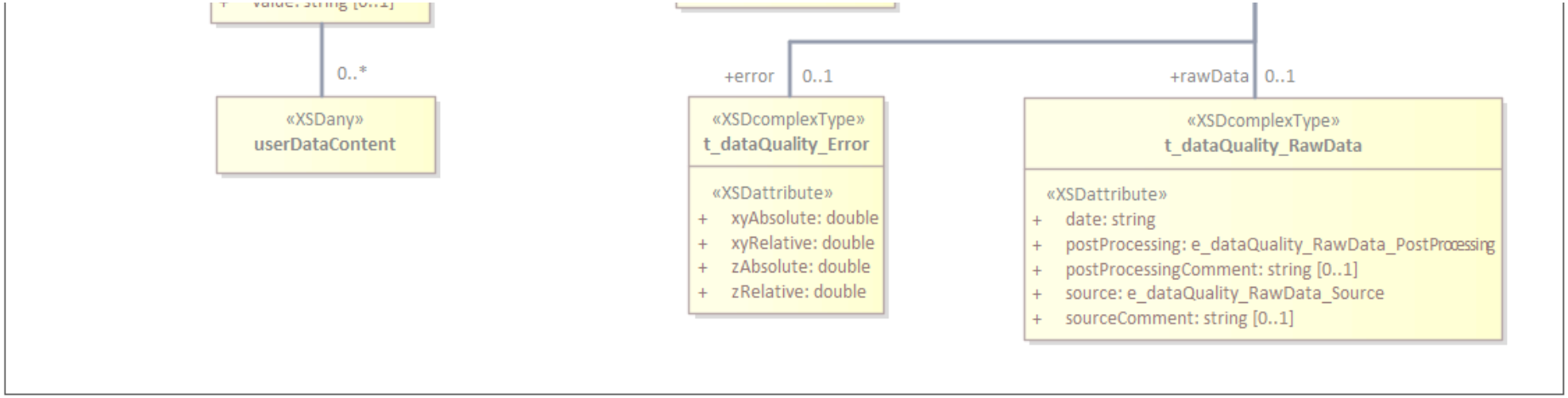


Figure 6. UML class diagram of the Core class

Figure 6 shows the UML class diagram of the ASAM OpenDRIVE Core class.

7.2 Additional user data

Additional or ancillary data contains data that are not yet described in ASAM OpenDRIVE, or data that is needed by an application for a specific reason, for example different road textures. It should be described near the element it refers to. They may be stored at any element in ASAM OpenDRIVE.

`<userData>` element

In ASAM OpenDRIVE, additional or ancillary data is represented by `<userData>` elements.

```
UML class: t_userData
XML tag:   <userData> (Multiplicity: 0..*)
```

Describes any additional information or data that is needed by an application for a specific reason.

Table 13. Attributes of the <userData> element

Name	Type	Use	Description
`code`	string	required	Code for the user data. Free text, depending on application.
`value`	string	optional	User data. Free text, depending on application.

7.3 Including additional data

Additional or external files can be included in the ASAM OpenDRIVE file. These files shall contain ASAM OpenDRIVE elements.

`<include>` element

In ASAM OpenDRIVE, additional files are represented by `<include>` elements.

```
UML class: t_include
XML tag:   <include> (Multiplicity: 0..*)
```

Provides information about additional files that should be included while processing the ASAM OpenDRIVE file.

Table 14. Attributes of the <include> element

Name	Type	Use	Description
`file`	string	required	Location of the file that is to be included

Multiple files can be combined with an `<include>` element at the appropriate locations. Upon parsing this element, ASAM OpenDRIVE readers shall immediately start reading the file specified as attribute of the element. It is the user's responsibility to make sure that contents read from an include file are consistent with the context from which the inclusion starts.

The parent element under which the `<include>` element occurs must be present in both, the parent file and the included file.

XML example

Original File

Code 1. `<planView>` element with the included file `planview.xml`

```
<planView>
    <include file="planview.xml"/>
</planView>
```

Included File

Code 2. Content of the file `planview.xml` with the `<planView>` element as root element

```
<planView>
    <geometry x="-0.014" y="-0.055" hdg="2.88" length="95.89" s="0.0">
        <arc curvature="-0.000490572"/>
    </geometry>
    <geometry x="-92.10" y="26.64" hdg="2.84" length="46.65" s="95.89">
        <spiral curvStart="-0.000490572" curvEnd="-0.004661241"/>
    </geometry>
</planView>
```

7.4 Description of the data quality

Raw data or data from external sources that is integrated in ASAM OpenDRIVE may be of varying quality. It is possible to describe quality and accuracy of external data in ASAM OpenDRIVE. They may be stored at any position in ASAM OpenDRIVE.

`<dataQuality>` element

In ASAM OpenDRIVE, data quality from external sources is represented by `<dataQuality>` elements.

```
UML class: t_dataQuality
XML tag:   <dataQuality> (Multiplicity: 0..1)
```

Describes the quality and accuracy of measurement data that is integrated into the ASAM OpenDRIVE file.

Measurement data derived from external sources like GPS that is integrated in ASAM OpenDRIVE may be inaccurate.

`<error>` element

In ASAM OpenDRIVE, the absolute or relative errors of road data are described by `<error>` elements within the `<dataQuality>` element.

```
UML class: t_dataQuality_Error
XML tag:   <error> (Multiplicity: 0..1)
```

Describes the error range, given in [m], of measurement data that is integrated into the ASAM OpenDRIVE file.

Table 15. Attributes of the <error> element

Name	Type	Use	Unit	Description
`xyAbsolute`	double	required	m	Absolute error of the road data in x/y direction
`xyRelative`	double	required	m	Relative error of the road data between adjacent element entries in x/y direction.
`zAbsolute`	double	required	m	Absolute error of the road data in z direction
`zRelative`	double	required	m	Relative error of the road data between adjacent element entries in z-direction.

Additional meta data about the raw data that is integrated can be stored in the ASAM OpenDRIVE.

`<rawData>` element

In ASAM OpenDRIVE, the raw data is described by the `<rawData>` element within the `<dataQuality>` element.

```
UML class: t_dataQuality_RawData
XML tag:   <rawData> (Multiplicity: 0..1)
```

Describes some basic metadata containing information about the raw data.

Table 16. Attributes of the <rawData> element

Name	Type	Use	Description
`date`	string	required	Date of the delivery of raw data, to be given in ISO 8601 notification (YYYY-MM-DDTHH:MM:SS). Time-of-day may be omitted
`postProcessingComment`	string	optional	Comments concerning the post processing attribute. Free text, depending on the application
`postProcessing`	e_dataQuality_RawData_PostProcessing	required	Information about the kind of data handling before exporting data into the ASAM OpenDRIVE file
`sourceComment`	string	optional	Comments concerning the @source . Free text, depending on the application
`source`	e_dataQuality_RawData_Source	required	Source that has been used for retrieving the raw data; further sources to be added in upcoming versions